

GRAIL

NASDAQ : GRAL ·
Young-Company DCF
(Damodaran)

\$3.1B mkt cap · 44M sh · \$823M cash, zero debt · \$0.823B cash, no debt; ~43M shares (low float); Illumina holds ~12.5%;
Revenue-generating: \$147M FY25 (+17%), Galleri self-pay; burn ~\$300M/yr (~2.5yr); NHS-Galleri missed primary endpoint
(Feb'26); FDA PMA filed; Medicare law signed

\$73.00

THE CENTRAL QUESTION

Does ~\$3B / ~20x sales price GRAIL's multi-gate regulatory option fairly — after the NHS-Galleri miss, but with the FDA filing and Medicare law in hand?

GRAIL trades at ~\$3B (~\$73/sh on just ~43M shares) on ~\$147M FY25 revenue and ~\$823M cash. Galleri — a blood test for 50+ cancers, self-pay today — is a multi-gate regulatory option: FDA PMA (filed Jan 2026) → Medicare coverage (signed Feb 2026; ≥2028) → broad adoption. The shape is unusual: the pivotal NHS-Galleri trial MISSED its endpoint (-50% stock), but the FDA filing rests on the positive PATHFINDER 2 and the Medicare law passed — partly broken, partly de-risked. The finding: at \$73 the market roughly prices the modal regulatory-success case, with FDA-rejection downside and broad-adoption upside about balanced.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$79.54
expected fair value today
+9.0% vs spot \$73.00

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$123.72	\$192.59	\$300.05	\$467.89
1.69× spot	2.64× spot	4.11× spot	6.41× spot

WEIGHTING RATIONALE

Ultra Bear 20% FDA rejects; no coverage; near-total loss.

Bear 28% FDA late; Medicare narrow; sub-scale; low teens.

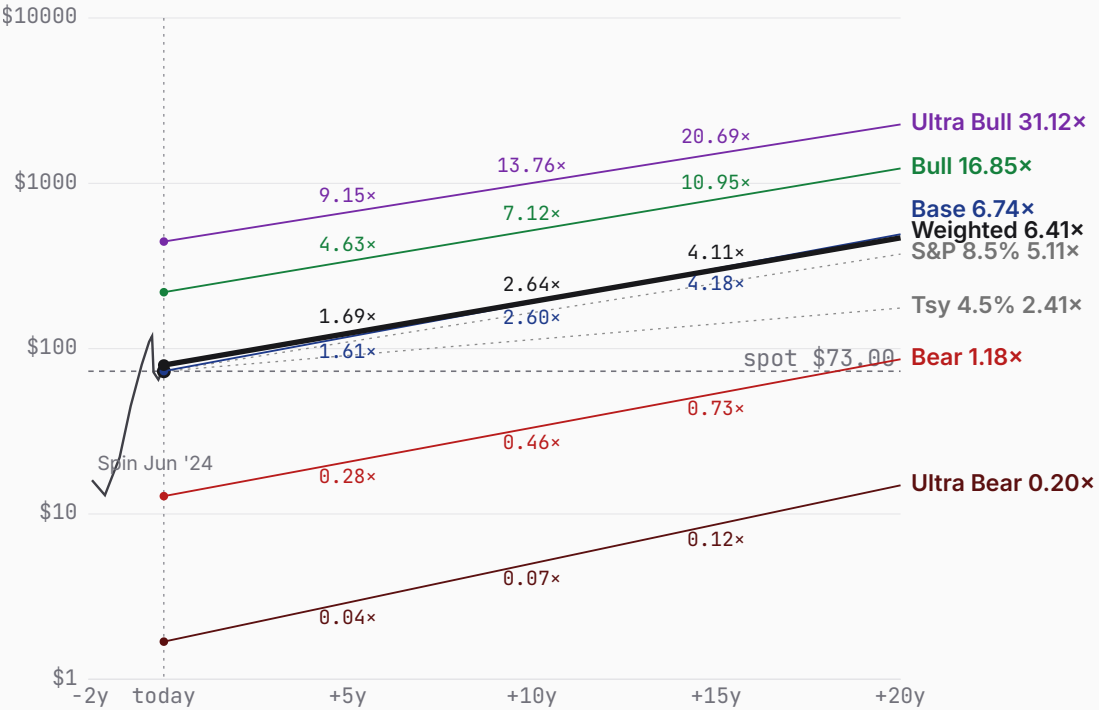
Base 34% FDA 2027 + Medicare 2028; leader; ≈ spot.

Bull 13% Broad adoption; category leader; ~3×.

Ultra Bull 5% MCED standard of care; dominant; ~6×.

Bull (13%) + Ultra Bull (5%) together contribute \$50.75 — 64% of the \$79.54 expected value despite 18% combined probability. Today's spot (\$73.00) sits 8% below the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$1.69	BEAR	\$12.82	BASE	\$73.13	BULL	\$219.48	ULTRA BULL	\$444.35
	-97.7% vs spot		-82.4% vs spot		+0.2% vs spot		+200.7% vs spot		+508.7% vs spot
TAM 1.3% P(fail) 40% S2C 1.0 Prob 20%		TAM 4.1% P(fail) 22% S2C 1.1 Prob 28%		TAM 10.4% P(fail) 12% S2C 1.3 Prob 34%		TAM 18.0% P(fail) 6% S2C 1.5 Prob 13%		TAM 27.2% P(fail) 3% S2C 1.7 Prob 5%	
FDA rejects; no coverage; self-pay niche.		FDA late; Medicare narrow; modest volume.		FDA 2027 + Medicare 2028; category leader.		Broad adoption; annual-repeat screening.		MCED standard of care; GRAIL dominant.	



GRAIL scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 20% Bear 28% Base 34% Bull 13% Ultra Bull 5% · Weighted expected \$79.54 (+9.0% vs spot)

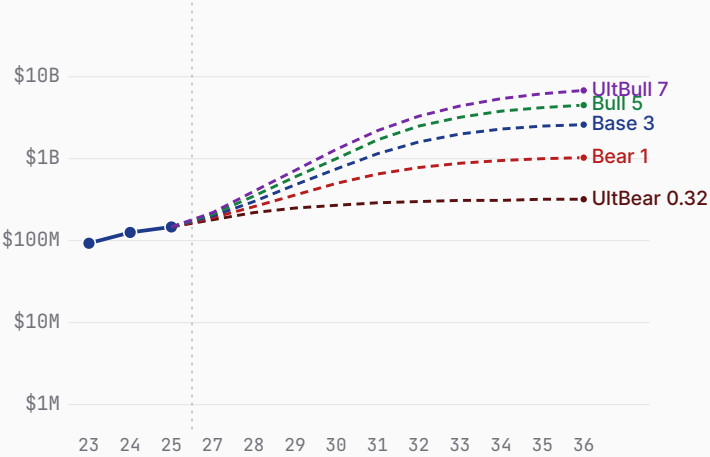
ULTRA BEAR	\$1.69	BEAR	\$12.82	BASE	\$73.13	BULL	\$219.48	ULTRA BULL	\$444.35
	-97.7% vs spot		-82.4% vs spot		+0.2% vs spot		+200.7% vs spot		+508.7% vs spot
Probability 20%		Probability 28%		Probability 34%		Probability 13%		Probability 5%	
FDA rejects; no coverage; self-pay niche.		FDA late; Medicare narrow; modest volume.		FDA 2027 + Medicare 2028; category leader.		Broad adoption; annual-repeat screening.		MCED standard of care; GRAIL dominant.	
WHY 20%		WHY 28%		WHY 34%		WHY 13%		WHY 5%	
The NHS-Galleri miss is a real, fired datapoint — the only RCT of any MCED failed its stage-shift primary endpoint, hardening FDA and payer skepticism. 40% within-scenario failure reflects genuine PMA-rejection risk. 20% scenario weight.		FDA approves but adoption disappoints — Medicare narrow, private payers await mortality proof, ASP compresses. 22% failure probability. 28% weight as a very plausible 'approved but sub-scale' outcome.		Requires FDA approval (on PATHFINDER 2, which was positive) + Medicare activation (the law is already signed) + broad coverage — the machinery is in motion. 12% failure probability. 34% weight as the central, modal outcome.		A chain: clean FDA + Medicare + broad private coverage + annual-repeat adoption + category leadership. Each plausible at 50-70%; jointly ~13%. GRAIL's breadth and only-RCT dataset are the edge.		Tail of tails: MCED standard of care + eventual mortality proof + GRAIL dominance + premium terminal. Individually 40-60%; jointly ~5%. The asymmetric upside the ~43M share count amplifies.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
The FDA PMA is rejected or stalled — the NHS-Galleri primary-endpoint miss (no significant stage III+IV reduction) hardens FDA/USPSTF skepticism about clinical utility, and Galleri stays a self-pay LDT niche. Test volume plateaus, the Medicare benefit never activates (it is gated on FDA approval), and GRAIL burns its ~\$823M cash defending the franchise.		The FDA approves but late, and Medicare coverage arrives narrow and slow; private payers wait for the mortality data the NHS trial did not deliver. Galleri reaches modest reimbursed volume (~\$1B revenue by 2035) at thin-but-improving margins — a real test, sub-scale economics.		The modal path: the FDA approves Galleri (~2027) on the PATHFINDER 2 evidence, Medicare coverage activates (~2028), and broad private coverage follows — Galleri becomes the leading reimbursed MCED test at ~\$2.6B revenue by 2035 (~10% of a ~\$25B screening market) at ~30% operating margins.		The bull: FDA + Medicare clear cleanly, broad private coverage and annual-repeat screening take hold, and GRAIL extends its first-mover, broadest-test, only-RCT-dataset lead — ~\$4.5B revenue by 2035 at ~36% operating margins. The screening category compounds and GRAIL leads it.		The tail: MCED becomes standard of care, the extended NHS follow-up eventually shows a mortality benefit, and GRAIL is the dominant platform — ~\$6.8B revenue by 2035 at ~40% margins with a premium terminal.	
With a 40% within-scenario failure probability and revenue stalled near \$0.3B, the equity collapses toward the cash-and-distress floor — a near-total loss from today's \$73.		The DCF lands in the low teens — well below spot. This is the 'approved, but adoption disappoints' world the NHS miss makes plausible.		DCF ~\$82/share — roughly today's price. The market is pricing this modal regulatory-success case; the small ~50M share count makes the per-share value swing hard on each gate.		DCF ~\$232/share — ~3× today's price. This is the scenario the regulatory machinery (PMA filed + Medicare law) keeps alive despite the NHS miss.		DCF ~\$457/share — ~6× spot. The asymmetric upside the low share count amplifies; individually each gate is plausible, jointly ~5%.	

GRAIL business snapshot

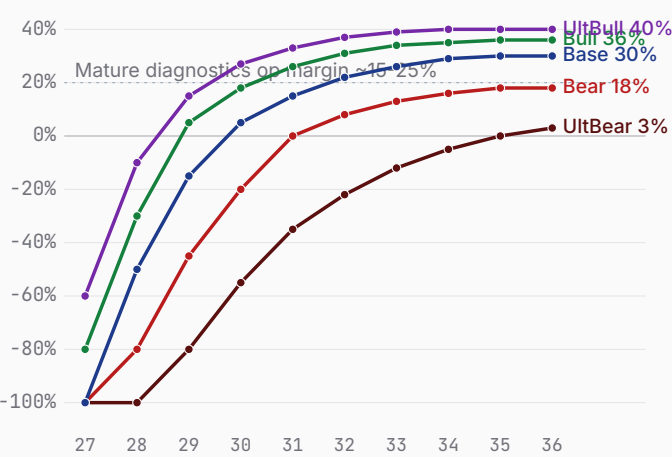
Bear \$12.82 Base \$73.13 Bull \$219.48

FY26-FY35 scenario projections (revenue-generating; pre-profit) · fiscal years end Dec 31 · Q1 2026 10-Q, FY25 results

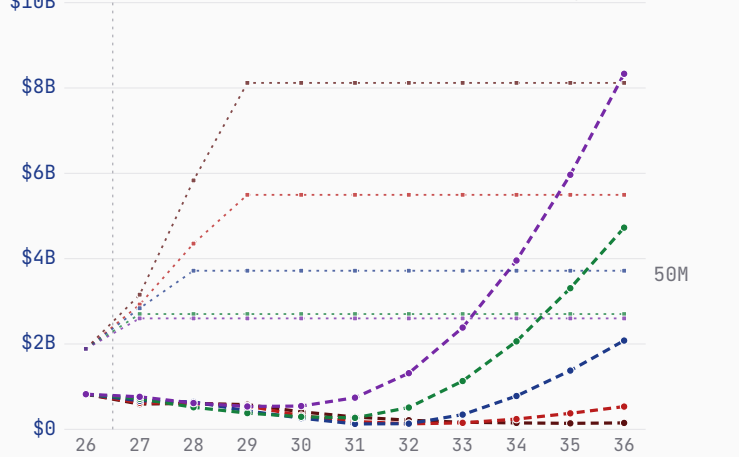
Revenue — history + scenarios to FY36 (log)



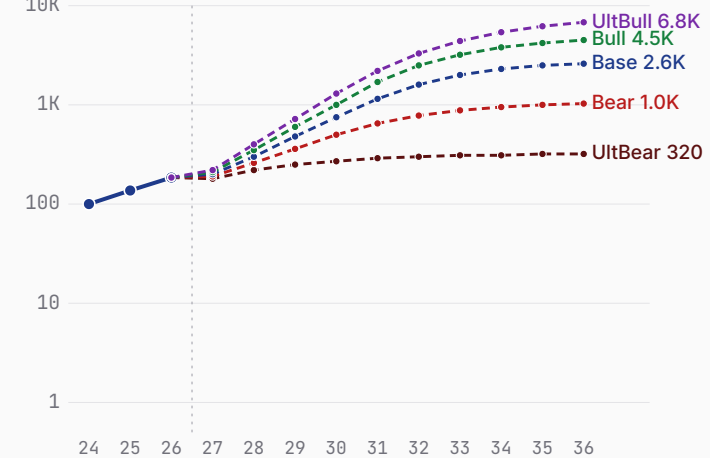
Path to profitability — op margin (FY27→FY36)



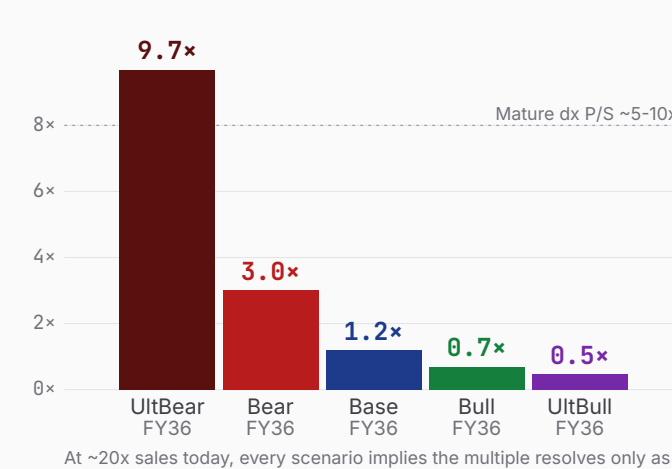
Cash + dilution (FY26 → FY36)



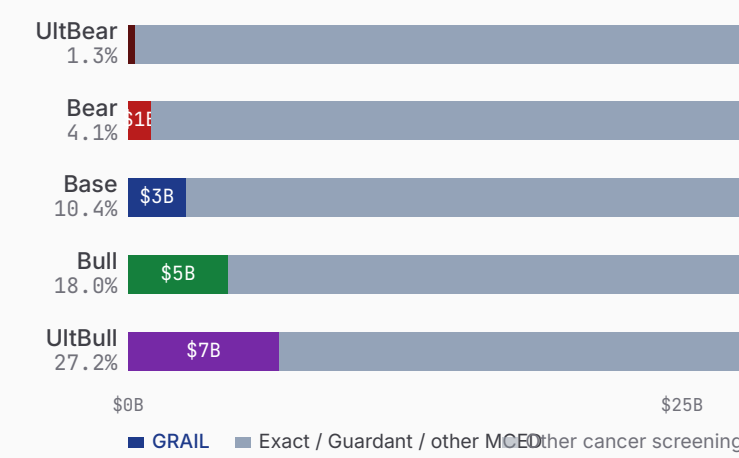
Galleri tests sold (000s/yr, log)



\$3.10B mkt cap as P/S on FY36 rev



\$25B FY35 MCED screening TAM — GRAIL share



Sources: GRAIL Q1 2026 10-Q (cash \$0.823B), FY2025 results (\$147M revenue, net loss), NHS-Galleri (primary endpoint missed Feb'26) + PATHFINDER 2 (positive) readouts, FDA PMA filing, Medicare MCED Coverage Act. Revenue modeled as reimbursed test volume x ASP; FCF NOL-shielded pre-tax over the window (documented simplification). TAM: MCED screening serviceable slice, ~\$25B annual by FY35.

GRAIL 7 Powers · the moat, scored

Bear \$12.82 Base \$73.13 Bull \$219.48

Arena. Multi-cancer early detection (MCED) blood screening — GRAIL (Galleri, 50+ cancers, only RCT) vs Exact Sciences (Cancerguard + Cologuard cash engine), Guardant (Shield, FDA-approved CRC → MCED), Freenome, Delfi; the race to originate the first FDA-approved, reimbursed MCED franchise.

POWER ORIENTATION
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	A reimbursed-volume cost curve is future; today's self-pay volume is sub-scale.
Network Economies	No network flywheel yet; data scale is the closest analog.
Counter-Positioning	Galleri's 50-cancer breadth + build-own dataset is hard for single-cancer (Shield CRC) entrants to mirror without their own large trials.
Switching Costs	Provider-ordering and EHR integration build mild lock-in; early.
Branding	The MCED category-definer brand; strong but dented by the NHS miss.
Cornered Resource · thesis leans here	The only randomized MCED dataset (NHS ~140k) + first PMA + largest commercial test base — the strongest, if contested, origination asset.
Process Power	Assay + bioinformatics pipeline is real but not a demonstrated durable process edge.

THE RACE · NAMED RIVALS

Exact Sciences	~\$3B-rev co.; self-funded
~25% terminal share · Cancerguard MCED (LDT, launched 2025) funded by the profitable Cologuard franchise; huge commercial muscle.	
Guardant Health	~\$700M-rev co.
~22% terminal share · Shield FDA-approved for CRC (Medicare \$1,495); extending to MCED (Breakthrough Device).	
Freenome	\$1.35B private
~10% terminal share · CRC-first → multi-cancer; ~\$1.35B raised (Roche, T. Rowe).	
Delfi Diagnostics	Merck GHI + rounds
~5% terminal share · FirstLook Lung (fragmentomics); niche, lung-first.	

LEAD / LAG · FALSIFIERS

MCED clinical evidence (RCT)	NHS ~140k (only RCT; primary missed) vs none randomized	LEADING
FDA approval	PMA filed (decision 2026-27) vs Guardant Shield approved (CRC)	ROUGHLY LEVEL
Reimbursement / commercial scale	self-pay; Medicare gated to 2028 vs Exact Cologuard reimbursed engine	LAGGING

COMPETITIVE READ

GRAIL is originating the MCED category — the broadest test (50+ cancers), the only randomized dataset, and the first PMA — but the NHS-Galleri primary-endpoint miss dented the 'proven population benefit' narrative, and Exact (Cologuard cash engine) and Guardant (FDA + Medicare in CRC) are scaling MCED behind it. The origination window is open but closing: GRAIL must convert the PMA into approval + Medicare coverage before better-capitalized rivals scale, and the extended NHS mortality follow-up is the falsifier that could reopen — or close — the whole thesis.

GRAIL show your work

Bear \$12.82 Base \$73.13 Bull \$219.48 · Weighted \$79.54 (+9.0%)

ULTRA BEAR					\$1.69	BEAR					\$12.82	BASE					\$73.13	BULL					\$219.48	ULTRA BULL					\$444.35
ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS					
TAM Year-10 (\$B)					25	TAM Year-10 (\$B)					25	TAM Year-10 (\$B)					25	TAM Year-10 (\$B)					25	TAM Year-10 (\$B)					25
Mature share (%)					1.3	Mature share (%)					4.1	Mature share (%)					10.4	Mature share (%)					18.0	Mature share (%)					27.2
Mature op margin (%)					3	Mature op margin (%)					18	Mature op margin (%)					30	Mature op margin (%)					36	Mature op margin (%)					40
Sales-to-capital					1.0	Sales-to-capital					1.1	Sales-to-capital					1.3	Sales-to-capital					1.5	Sales-to-capital					1.7
Terminal WACC (%)					11.5	Terminal WACC (%)					10.0	Terminal WACC (%)					10.0	Terminal WACC (%)					9.0	Terminal WACC (%)					8.5
Terminal growth (%)					2.0	Terminal growth (%)					2.5	Terminal growth (%)					2.5	Terminal growth (%)					3.0	Terminal growth (%)					3.0
P(failure) (%)					40	P(failure) (%)					22	P(failure) (%)					12	P(failure) (%)					6	P(failure) (%)					3
Scenario probability (%)					20	Scenario probability (%)					28	Scenario probability (%)					34	Scenario probability (%)					13	Scenario probability (%)					5
10-YEAR DCF · \$B						10-YEAR DCF · \$B						10-YEAR DCF · \$B						10-YEAR DCF · \$B						10-YEAR DCF · \$B					
FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF	
FY27	0.18	-140%	-0.28	-0.25		FY27	0.19	-120%	-0.27	-0.23		FY27	0.20	-100%	-0.24	-0.21		FY27	0.21	-80%	-0.21	-0.19		FY27	0.22	-60%	-0.17	-0.16	
FY28	0.22	-110%	-0.28	-0.21		FY28	0.26	-80%	-0.27	-0.21		FY28	0.30	-50%	-0.23	-0.18		FY28	0.35	-30%	-0.20	-0.16		FY28	0.40	-10%	-0.15	-0.12	
FY29	0.25	-80%	-0.23	-0.15		FY29	0.36	-45%	-0.25	-0.17		FY29	0.48	-15%	-0.21	-0.15		FY29	0.60	+5%	-0.14	-0.10		FY29	0.72	+15%	-0.08	-0.06	
FY30	0.27	-55%	-0.17	-0.10		FY30	0.50	-20%	-0.23	-0.14		FY30	0.75	+5%	-0.17	-0.11		FY30	1.00	+18%	-0.09	-0.06		FY30	1.30	+27%	+0.01	+0.01	
FY31	0.29	-35%	-0.12	-0.06		FY31	0.65	+0%	-0.14	-0.07		FY31	1.15	+15%	-0.14	-0.07		FY31	1.70	+26%	-0.03	-0.01		FY31	2.20	+33%	+0.20	+0.12	
FY32	0.30	-22%	-0.08	-0.04		FY32	0.78	+8%	-0.06	-0.03		FY32	1.60	+22%	+0.01	+0.00		FY32	2.50	+31%	+0.24	+0.13		FY32	3.30	+37%	+0.57	+0.31	
FY33	0.31	-12%	-0.05	-0.02		FY33	0.88	+13%	+0.02	+0.01		FY33	2.00	+26%	+0.21	+0.10		FY33	3.20	+34%	+0.62	+0.29		FY33	4.40	+39%	+1.07	+0.53	
FY34	0.31	-5%	-0.01	-0.01		FY34	0.95	+16%	+0.09	+0.03		FY34	2.30	+29%	+0.44	+0.18		FY34	3.80	+35%	+0.93	+0.40		FY34	5.40	+40%	+1.57	+0.72	
FY35	0.32	+0%	-0.01	-0.00		FY35	1.00	+18%	+0.14	+0.05		FY35	2.50	+30%	+0.60	+0.22		FY35	4.20	+36%	+1.25	+0.49		FY35	6.20	+40%	+2.01	+0.84	
FY36	0.32	+3%	+0.01	+0.00		FY36	1.03	+18%	+0.16	+0.05		FY36	2.60	+30%	+0.70	+0.24		FY36	4.50	+36%	+1.42	+0.51		FY36	6.80	+40%	+2.37	+0.91	
Σ PV explicit FCF					-0.84	Σ PV explicit FCF					-0.71	Σ PV explicit FCF					+0.02	Σ PV explicit FCF					+1.31	Σ PV explicit FCF					+3.11
+ PV terminal (g 2.0%)					0.03	+ PV terminal (g 2.5%)					0.70	+ PV terminal (g 2.5%)					3.24	+ PV terminal (g 3.0%)					8.79	+ PV terminal (g 3.0%)					17.10
EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE					
Operating EV					\$-0.81B	Operating EV					\$-0.01B	Operating EV					\$3.26B	Operating EV					\$10.10B	Operating EV					\$20.21B
+ Cash & investments					\$0.82B	+ Cash & investments					\$0.82B	+ Cash & investments					\$0.82B	+ Cash & investments					\$0.82B	+ Cash & investments					\$0.82B
= Equity value					\$0.01B	= Equity value					\$0.81B	= Equity value					\$4.08B	= Equity value					\$10.92B	= Equity value					\$21.03B
Raised over period					\$0.70B	Raised over period					\$0.65B	Raised over period					\$0.40B	Raised over period					\$0.20B	Raised over period					\$0.20B
Dilution by FY36					+50%	Dilution by FY36					+27%	Dilution by FY36					+14%	Dilution by FY36					+7%	Dilution by FY36					+5%
FY36 shares (M)					66	FY36 shares (M)					56	FY36 shares (M)					50	FY36 shares (M)					47	FY36 shares (M)					46
DCF per share					\$0.15	DCF per share					\$14.46	DCF per share					\$81.60	DCF per share					\$232.34	DCF per share					\$457.17
× (1-P_fail) [60%]					\$0.09	× (1-P_fail) [78%]					\$11.28	× (1-P_fail) [88%]					\$71.81	× (1-P_fail) [94%]					\$218.40	× (1-P_fail) [97%]					\$443.45
+ P_fail × distress [40% × \$4.00]					\$1.60	+ P_fail × distress [22% × \$7.00]					\$1.54	+ P_fail × distress [12% × \$11.00]					\$1.32	+ P_fail × distress [6% × \$18.00]					\$1.08	+ P_fail × distress [3% × \$30.00]					\$0.90
= Expected per share					\$1.69	= Expected per share					\$12.82	= Expected per share					\$73.13	= Expected per share					\$219.48	= Expected per share					\$444.35
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$2.91	\$5.02	\$8.65	\$14.91		\$20.65	\$33.25	\$53.55	\$86.25		\$117.78	\$189.68	\$305.48	\$491.98		\$337.70	\$519.59	\$799.45	\$1,230.06		\$668.15	\$1,004.67	\$1,510.68	\$2,271.54		\$668.15	\$1,004.67	\$1,510.68	\$2,271.54	
0.04×	0.07×	0.12×	0.20×		0.28×	0.46×	0.73×	1.18×		1.61×	2.60×	4.18×	6.74×		4.63×	7.12×	10.95×	16.85×		9.15×	13.76×	20.69×	31.12×		9.15×	13.76×	20.69×	31.12×	

GRAIL supporting analysis · disclaimers · glossary

Bear \$12.82 Base \$73.13 Bull \$219.48

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

Only randomized MCED dataset.
The NHS-Galleri RCT (~140k) is the largest, only randomized MCED dataset — a cornered resource even after the primary-endpoint miss (stage IV fell -14%; mortality follow-up extended).
- 2

FDA filing rests on PATHFINDER 2, not NHS.
The PMA is built on the positive PATHFINDER 2 (69.8% sensitivity for the 12 deadliest, 99.6% specificity) plus the NHS prevalent round — not the failed late-stage primary.
- 3

Medicare law already signed.
The MCED coverage benefit is law (Feb 2026); coverage is gated on FDA approval and earliest 2028, but the legislative risk is retired.
- 4

Cash-funded into the coverage window.
~\$823M and burn halved to ~\$300M/yr funds GRAIL into the ~2028 Medicare window without an imminent raise.
- 5

Tiny share count amplifies the unlock.
~43M shares means a reimbursed-volume inflection moves per-share value hard — the asymmetry cuts both ways.

FALSIFICATION TRIGGERS

Bull validation FDA PMA approval (2026-27) · Medicare NCD activates (~2028) · broad private coverage · test volume sustains >40% growth	Bear validation PMA rejected/delayed · Medicare coverage narrow or post-2028 · private payers demand mortality data	Reframe needed if the extended NHS follow-up shows a mortality benefit, the clinical-utility debate resolves and the bull/ultra weights rise materially
---	---	---

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice. This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.	Not a professional. The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.	AI-assisted analysis. Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.
Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$GRAL or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Galleri — GRAIL's blood test screening for 50+ cancers from one draw; sold self-pay (~\$949) today as an LDT, FDA PMA filed Jan 2026.	MCED — Multi-cancer early detection — blood-based screening across many cancer types; a new category GRAIL pioneered.	NHS-Galleri — The ~140k-participant UK randomized trial; missed its stage-shift primary endpoint (Feb 2026) but cut stage IV; mortality follow-up extended.
PMA / NCD — FDA Premarket Approval (filed Jan 2026) and Medicare National Coverage Determination (earliest 2028) — the two gates to reimbursed scale.	Sales-to-capital (s2c) — Damodaran reinvestment efficiency — incremental revenue per dollar of reinvested capital.	